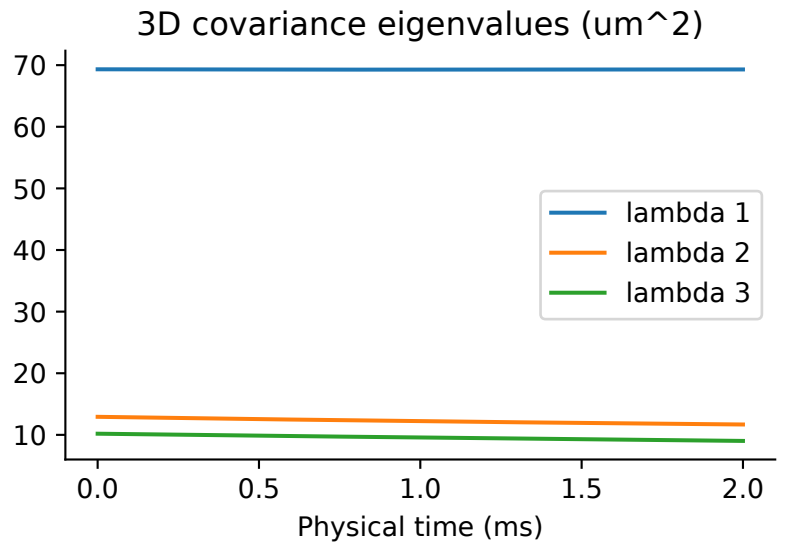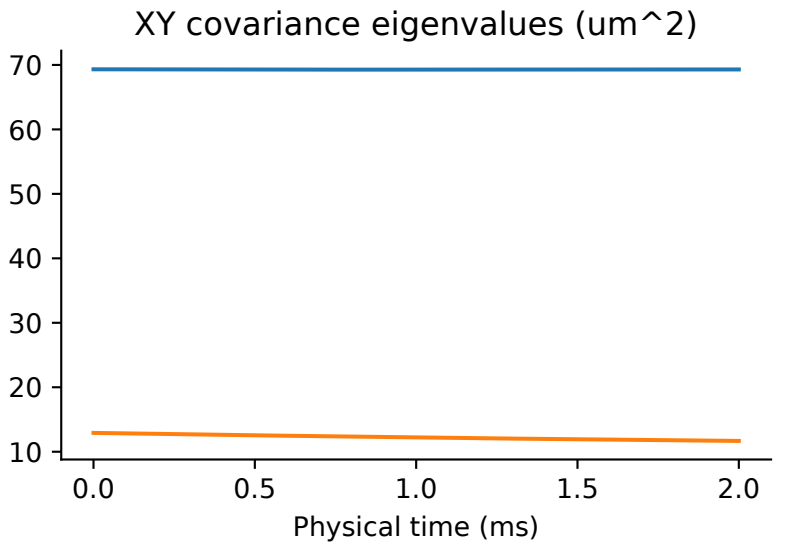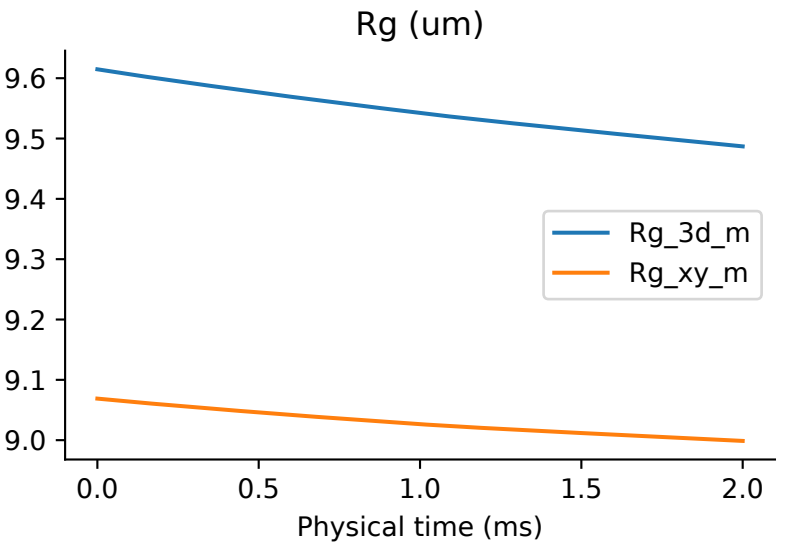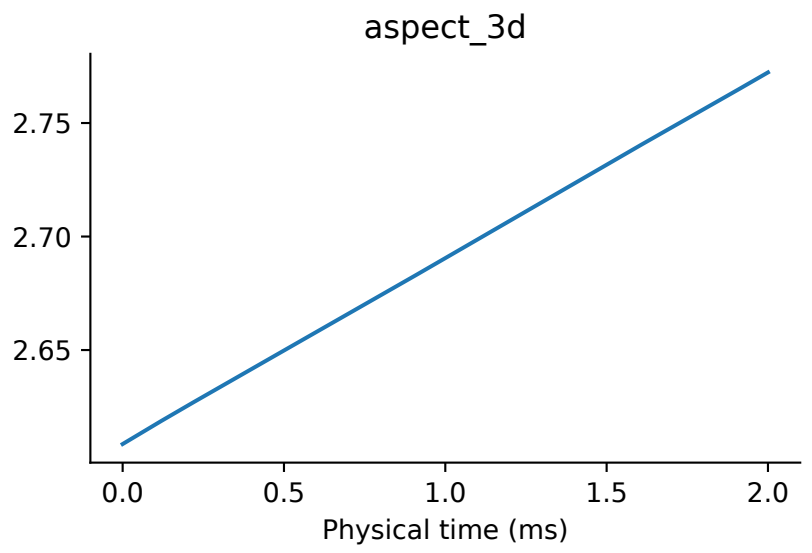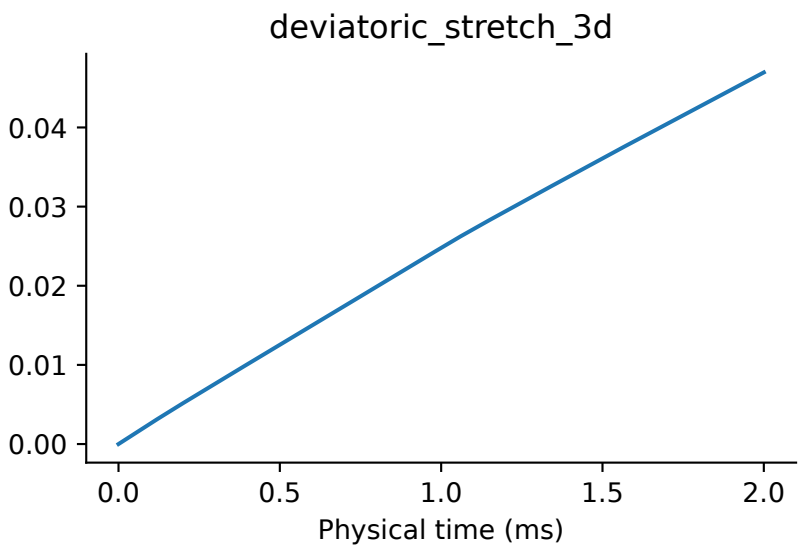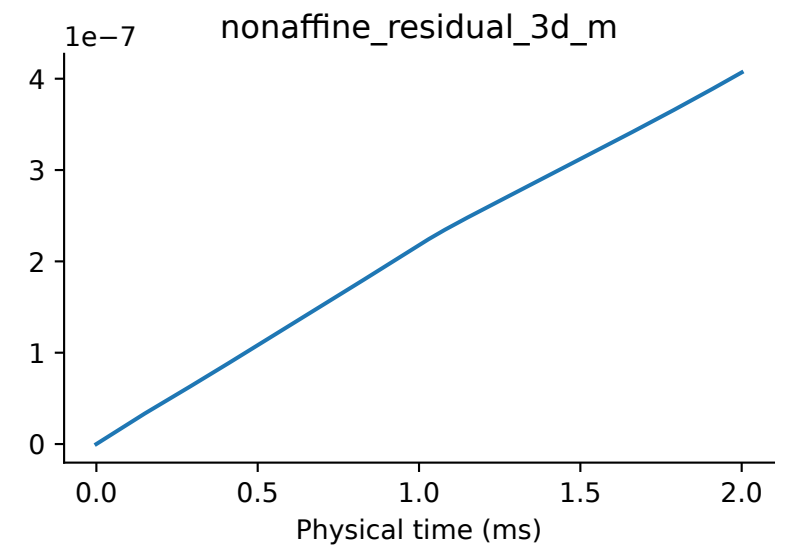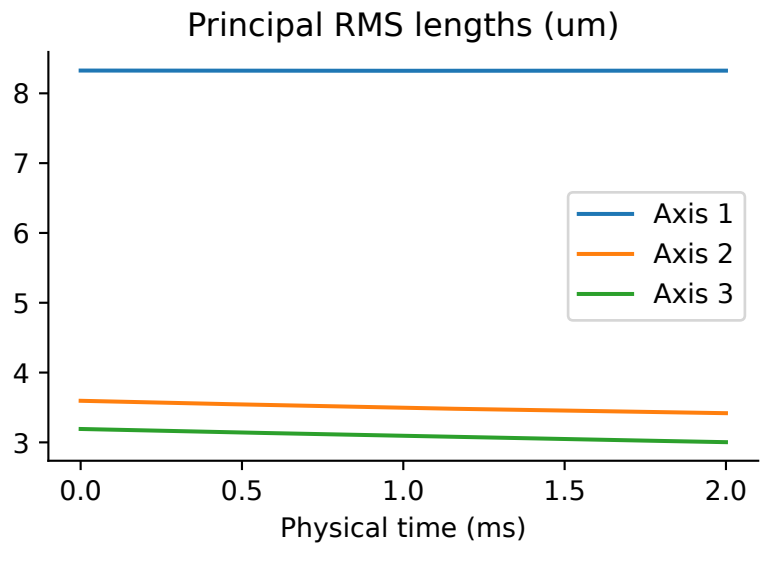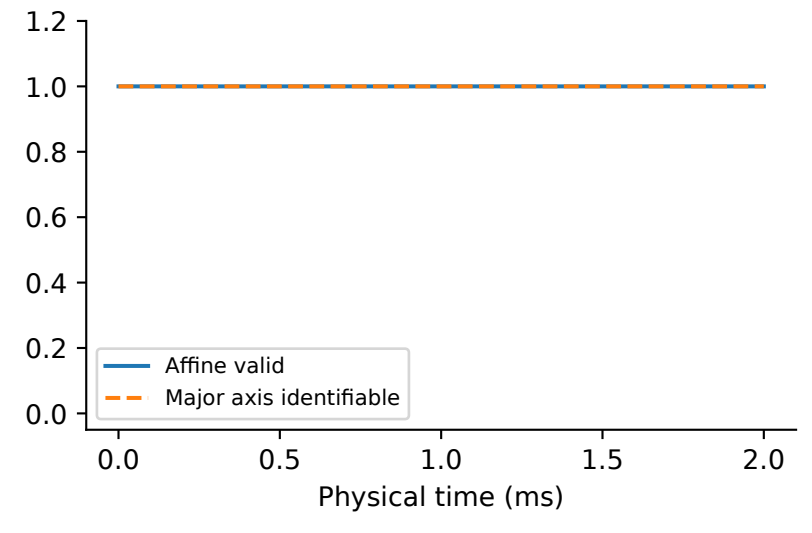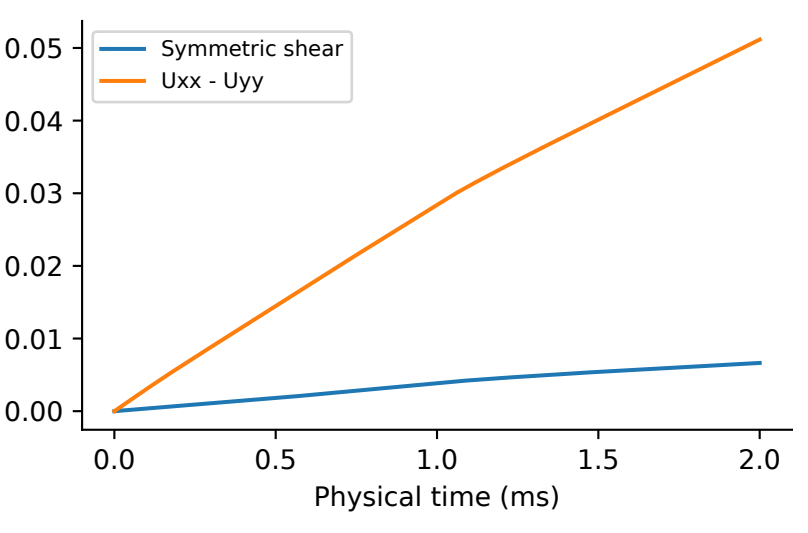Covariance, anisotropy and non-affine evolution